The prime factorisation of a triple product of gcds, and an off-by-one in the conjecture as posted

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Abstract

OEIS A129454 is $a(n) = \prod_{i=1}^{n-1}\prod_{j=1}^{n-1}\prod_{k=1}^{n-1}\gcd(i,j,k)$. In April 2007 P. Bala conjectured that $\operatorname{ord}_p a(n) = \sum_{t\geq 1}\lfloor n/p^t\rfloor^3$. We prove the exact prime factorisation $\operatorname{ord}_p a(n) = \sum_{t\geq 1}\lfloor (n-1)/p^t\rfloor^3$, by the counting argument that also gives Legendre's formula. The conjecture as literally indexed is false — the comment was transplanted from the companion entry A092287, whose products run to n rather than $n-1$, and the upper limit was not adjusted. The smallest counterexample is $n = 4$, $p = 2$: the posted formula gives exponent 9, the truth is 1, and indeed $a(4) = 6$. With n replaced by $n-1$ throughout, the conjecture is exactly right, and that is the reading the entry's own definition forces.

1 The sequence and the conjecture

OEIS A129454, contributed by Peter Bala on 16 April 2007, has offset 0 and

$$a(n) = \prod_{i=1}^{n-1}\prod_{j=1}^{n-1}\prod_{k=1}^{n-1}\gcd(i,j,k) \quad (n > 2), \qquad a(n) = 1 \ (n \leq 2),$$

with $a(0), \ldots, a(7) = 1, 1, 1, 2, 6, 1536, 7680, 8806025134080$. The entry carries

> **Conjecture:** Let p be a prime and let $\operatorname{ord}_p(n,p)$ denote the largest power of p which divides n. ... Then we conjecture that the prime factorization of $a(n)$ is given by $\operatorname{ord}_p(a(n),p) = (\lfloor n/p\rfloor)^3 + (\lfloor n/p^2\rfloor)^3 + (\lfloor n/p^3\rfloor)^3 + \cdots$. Compare with the comments in A092287.

As of the "Last modified" line on the live entry (23 August 2026) this is still recorded as an unproven conjecture, with no proof in the entry's links or programs.

The pointer to A092287 explains the shape of the statement and also its defect. In A092287 the products run over $1 \leq j, k \leq n$ and the correct exponent is $\sum_t \lfloor n/p^t\rfloor^2$; here the products run over $1 \leq i, j, k \leq n-1$, but the comment reuses $\lfloor n/p^t\rfloor$ unchanged. As stated it is therefore false. Section 3 proves the corrected version and Section 4 records the counterexample.

Throughout p is a prime and $v_p(N)$ is the exponent of p in N.

2 The counting lemma

Lemma 1 (layer cake). *For positive integers $x_1, \ldots, x_d$,*

$$v_p\big(\gcd(x_1, \ldots, x_d)\big) = \min_{1\leq t\leq d} v_p(x_t) = \#\big\{\, s \geq 1 : p^s \mid x_t \text{ for every } t \,\big\}.$$

Proof. The first equality is the description of gcd prime by prime. For the second, p^s divides every x_t exactly when $s \leq v_p(x_t)$ for every t, that is when $1 \leq s \leq \min_t v_p(x_t)$. □

3 The theorem

Theorem 2. *Let $N \geq 1$, $d \geq 1$ and put $G_d(N) = \prod_{x_1=1}^{N} \cdots \prod_{x_d=1}^{N} \gcd(x_1, \dots, x_d)$. Then for every prime p,*

$$v_p\big(G_d(N)\big) = \sum_{s \geq 1} \left\lfloor \frac{N}{p^s} \right\rfloor^d .$$

Proof. By additivity of v_p and Lemma 1,

$$v_p\big(G_d(N)\big) = \sum_{x_1,\dots,x_d \leq N} \#\{s \geq 1 : p^s \mid x_1, \dots, p^s \mid x_d\} = \sum_{s \geq 1} \#\{(x_1, \dots, x_d) \in [1, N]^d : p^s \mid x_t \ \forall t\},$$

a finite rearrangement of nonnegative terms. The d divisibility conditions are independent, so the inner count is $\big(\#\{x \leq N : p^s \mid x\}\big)^d = \lfloor N/p^s \rfloor^d$. $\square$

Corollary 3. *For A129454 and every prime p,*

$$\operatorname{ord}_p a(n) = \sum_{s \geq 1} \left\lfloor \frac{n-1}{p^s} \right\rfloor^3 .$$

This is Bala's conjecture with n replaced by $n-1$, and it is the statement forced by the entry's own definition of $a(n)$ as a product over $1 \leq i, j, k \leq n-1$.

Proof. Apply Theorem 2 with $d = 3$ and $N = n - 1$. For $n \leq 2$ the product is empty and both sides are 0. $\square$

4 The conjecture as posted is false

Proposition 4. *The formula $\operatorname{ord}_p a(n) = \sum_{s \geq 1} \lfloor n/p^s \rfloor^3$ fails whenever $p \mid n$ and $n \geq 4$. The smallest failure is $n = 4$, $p = 2$.*

Proof. By Corollary 3 the truth is $\sum_s \lfloor (n-1)/p^s \rfloor^3$, and $\lfloor n/p^s \rfloor = \lfloor (n-1)/p^s \rfloor$ unless $p^s \mid n$, in which case $\lfloor n/p^s \rfloor = \lfloor (n-1)/p^s \rfloor + 1$. So if $p \mid n$ the two expressions differ, the posted one being strictly larger.

Explicitly at $n = 4$, $p = 2$: the only triples $(i, j, k) \in [1, 3]^3$ with $\gcd(i, j, k) > 1$ are $(2, 2, 2)$ and $(3, 3, 3)$, so $a(4) = 2 \cdot 3 = 6$ and $v_2(a(4)) = 1$. Corollary 3 gives $\lfloor 3/2 \rfloor^3 + \lfloor 3/4 \rfloor^3 = 1 + 0 = 1$, correct. The posted formula gives $\lfloor 4/2 \rfloor^3 + \lfloor 4/4 \rfloor^3 = 8 + 1 = 9$. Since $2^9 \nmid 6$, the posted formula is false at $n = 4$. No smaller n fails: for $n \leq 3$ every relevant $\lfloor n/p^s \rfloor$ agrees with $\lfloor (n-1)/p^s \rfloor$. $\square$

Remark (what to record on the entry)**.** The right correction is to replace n by $n - 1$ in the comment, not to change the definition: the definition, the DATA and the Mathematica, Magma and SageMath programs on the entry all agree on products over $1 \leq i, j, k \leq n - 1$. It is only the comment, transplanted from A092287, that is out of step.

Remark (honest assessment)**.** Theorem 2 is a three-line count and is the same argument that proves Legendre's formula; its $d = 2$ case is the rectangular statement that A092287 records. What is worth recording here is not the proof but the off-by-one: the statement people have been checking is not the statement that is true.

5 Verification

A verification script, written separately from this note, checks every claim and prints every number quoted. It (i) recomputes $a(0), \ldots, a(11)$ of A129454 directly from the triple product and matches the published DATA exactly; (ii) compares $v_p(a(n))$ against $\sum_s \lfloor (n-1)/p^s \rfloor^3$ for every $2 \leq n \leq 25$ and every prime $p < n$, with 0 mismatches; (iii) compares it against the posted formula $\sum_s \lfloor n/p^s \rfloor^3$ over the same range, finding 25 mismatches; and (iv) confirms that the smallest failure is $n = 4$, $p = 2$, where the true exponent is 1 and the posted formula claims 9, with $a(4) = 6 = 2 \cdot 3$.

References

[1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org/A129454, entry A129454, consulted 23 August 2026.

[2] ibid., entry A092287, https://oeis.org/A092287.

[3] A. M. Legendre, *Théorie des Nombres*, 3rd ed., Firmin Didot, Paris, 1830.